Lecture 08: Financial Analysis

Introduction

- We will work through the various ratios by looking at various Canadian P&C companies.
- Using the ratios, we will answer questions about the financial health of these companies.
- And we'll also understand if the ratios present any warning signs for companies.

Gross Written Premium / Operating DWP

- Gross Written Premium (GWP) and Operating Direct Written Premium (DWP) represent the total dollar amount of premiums written by an insurer during a specific period, before deducting any premiums ceded to reinsurance companies.
- This is the top-line revenue metric for an insurance business and reflects its market share and premium growth scale.
- When analyzing the metric, we are trying to understand the trajectory of the company's growth and the underlying forces shaping it.
  - The trend itself identifies whether the top-line premium volume is growing, stagnating, or declining.
  - We can also look at the breakdown by the lines of businesses and see if the growth is driven by a single booming product line or if it is balanced across the portfolio.
  - We can also connect the data to macro factors.
    - For example, if the premium volume increased because the insurer raised the premium rates due to high inflation or because of an economic boom, which increased the demand for commercial insurance.
  - We can also analyze if new regulations or intense competition forced companies to lower their premium prices.
- Remember the difference between written vs earned.
  - Written Premium represents the total value of all policies bound during that period, regardless of whether the cash has been collected. When a policy is written, the insurer secures the legal commitment for that revenue.
  - Earned premium, on the other hand, is the actual activity of the customer paying their premiums according to the contract or the insurance product

purchased. It contributes to the actual cash earnings which happen over a period of time.

- Note the slight nuance between GWP and Operating DWP:
  - Direct written premium refers strictly to the policies the company issued directly to its own customers via its own agents or brokers.
  - Gross written premium, on the other hand, is a broader umbrella term.
    - $\text{GWP} = \text{DWP} + \text{Assumed Premium}$
    - Assumed premium is money the company makes by acting as a re-insurer.
  - Basically, if an insurer only writes policies for everyday customers and does not act as a re-insurer for other companies, its DWP and GWP are exactly the same number.

Net Underwriting Revenue / Operating Net Underwriting Revenue

- Net underwriting revenue is the actual amount of premium money an insurance company gets to keep and recognizes as its revenue after paying for its safety net, which is reinsurance.
- $\text{Net Underwriting Revenue} = \text{Gross Premiums Earned} - \text{Premiums Paid to Reinsurers}$
- Net underwriting revenue includes everything related to premiums, even one-off accounting adjustments, oddities from discontinued product lines or unique regulatory payouts.
- On the other hand, operating net underwriting revenue strips away those one-offs and isolates the premium revenue generated strictly by the company's ongoing daily core insurance operations.
- Insurance companies treat premium as revenue because it is earned income for providing the service of taking on risk. The claims that may happen later are treated as expenses.
- ***If net underwriting revenue is significantly lower than GEP, it proves that a massive portion of the premiums and therefore the risk is being passed off to reinsurers.***

Underwriting Income (UW)

- This is the profit made strictly from the insurance operations.
- $\text{Underwriting Income} = \text{Net Underwriting Revenue} - \text{Claims} - \text{Operating Expenses}$
- If this number is negative, it means that the company paid out more in claims and expenses than it kept in premiums.

- If the underwriting income is fluctuating, check if specific product lines are driving the losses or gains.

Investment Income

- Insurance companies don't just let premium cash sit around while waiting for claims to be filed.
- They invest that cash into stocks, bonds, and real estate.
- This metric tracks the returns, dividends, and interest generated by that investment portfolio.
- When analyzing investment income, it is important to look at what their investment income is based on. Is the company heavily invested in safe, predictable government bonds, or are volatile market gains and losses swinging their bottom lines?

Investment Income vs Underwriting Income

- It is common to analyze the proportion of underwriting income vs investment income.
- Some companies are highly efficient insurers, making solid profits from their policies, which is reflected by a high UW income.
- Other companies can be mediocre insurers that rely entirely on a booming stock market to stay profitable, which is reflected by a high investment income.

Liquidity Measures

- Liquidity measures evaluate an insurer's immediate ability to pay out claims and short-term liabilities without needing to liquidate long-term investments at a loss.
- The two primary metrics are the cash ratio and the liabilities-to-liquid-assets ratio.

Cash Ratio

- $\text{Cash Ratio} = \frac{\text{Cash+Investment Income Due+Short-Term Deposits}}{\text{Net Current Liabilities}}$
- It basically measures the dollars of pure cash and near-cash assets existing for every dollar of immediate short-term debt and obligations due within a year.
- A higher cash ratio means higher short-term safety.
- Investment income due refers to investment income like bond interest or stock dividends that the insurance company has legally earned but has not yet received

in cash. It is an accounting receivable.

Liabilities to Liquid Assets Ratio

- It measures the proportion of total claims and obligations relative to assets that can be easily converted into cash within a short time frame.
- Lower the ratio, the safer it is, showing that liquid reserves easily cover potential obligations.

Cash Ratio vs Liabilities to Liquid Assets Ratio

Cash Ratio	Liabilities to Liquid Assets Ratio
Only counts pure cash or short-term deposits and investment income due.	Uses a much broader definition of liquid assets, which includes things like stocks and longer-term bonds that can be sold quickly if needed, even if they aren't cash today.
Only looks at current liabilities, which are obligations due within one year.	Often looks at total insurance liabilities, including long-term claims that might take years to pay out.

ROE

- ROE measures how effectively an insurer generates profits from shareholders' capital.
- $\text{ROE} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$
- It basically measures how much profit the insurer generates for every dollar of institutional and shareholder capital invested.
- A high ROE is ideal, but it must be cross-referenced with underwriting income. If ROE is high strictly because of volatile investment gains or taking on excessive underwriting risk, it may not be sustainable.

Debt-to-Equity Ratio

- It measures the mix of debt versus equity used to finance the firm's operations and growth.
- $\text{Debt-to-Equity} = \frac{\text{Total Debt}}{\text{Shareholders' Equity}}$
- Insurers typically rely less on traditional corporate debt compared to capital-intensive industries because their primary leverage comes from policyholder premiums and reserves.

- A high corporate debt-to-equity ratio on top of insurance liabilities can signal elevated insolvency risk during a bad claims year.

Minimum Capital Test (MCT)

- This is a risk-based solvency framework used by regulators to ensure insurers hold enough capital to absorb unexpected losses and to cover policyholder claims.
- $\text{MCT Ratio} = \frac{\text{Capital Available}}{\text{Minimum Capital Required}}$
- Capital available refers to Tier 1 equity, like common shares and retained earnings minus regulatory deductions.
- Minimum capital required is calculated by assessing the insurer's specific exposure across several risk categories, such as insurance risk, market risk, credit risk, and operational risk.
- In Canada, insurers must maintain specific thresholds (MCT Ratio):
  - 100% minimum requirement: Falling below this means potential insolvency or immediate regulatory intervention.
  - 150% supervisory target: The standard baseline regulators expect insurers to maintain to absorb market volatility without triggering early warning intervention.
  - Most well-managed companies set their own higher internal targets to ensure that they remain safely above 150% even after severe catastrophic years.

Policies In Force (PIF)

- This counts the total number of active insurance contracts an insurer has on its books at a single point in time.
- If revenue goes up by 10% because the company raised rates by 10%, PIF stays flat.
- On the other hand, if revenue goes up 10% because they sold 10% more policies, PIF increases by 10%, which indicates real market share growth.
- Comparing PIF from period to period helps to calculate customer retention, or dropping PIF alongside rising revenues signals that policyholders are fleeing rate hikes. This is a key warning sign for long-term growth.
- Average premium per policy shows whether the insurer targets high-value individual risks or high-volume, low-premium retail clients.
  - $\text{Average Premium per Policy} = \frac{\text{Total Written Premium}}{\text{Policies in Force}}$

Claims Ratio

- Claims ratio measures the percentage of earned premiums used to pay out claims and adjusting expenses.
- A high claims ratio points to poor risk selection, underpricing of policies, or an uptick in catastrophic events like storms or floods.
- $\text{Claims Ratio} = \frac{\text{Net Claims Incurred}}{\text{Net Underwriting Revenue}}$

Expense Ratio

- Expense ratio measures the operational overhead required to acquire and service policies.
- This reflects the administrative efficiency and scale.
- $\text{Expense Ratio} = \frac{\text{Underwriting Expenses}}{\text{Net Underwriting Revenue}}$

Combined Ratio

- Combined ratio measures the total operational outlays divided by the net premium revenue.
- It ignores investment income entirely to isolate pure underwriting performance.
- $\text{Combined Ratio} = \text{Claims Ratio} + \text{Expense Ratio}$
- If the combined ratio is equal to 95%, it signals efficient operation. The company keeps 5% of premiums as profit before even touching investment returns.
- If the combined ratio is 105%, it means underwriting loss. The company spent more on claims and overhead than it brought in via premiums, meaning that it relies heavily on its investment returns to turn a profit.

Expenses

- Insurance expenses are heavily scrutinized relative to premium volume to evaluate operational efficiency. They fall into two main categories:
  1. Acquisition costs
     - These are fees paid to brokers or agents who sell policies.
     - These also include marketing and distribution costs, such as advertising campaigns and direct sales costs to attract policyholders.
     - Premium taxes also fall under this category.
  2. General operating expenses

- These are expenses tied to evaluating risk, pricing policies, and setting up reserves.
- Also includes technology and infrastructure costs, such as core insurance platforms, policy management systems, and cyber security.
- Administrative overhead, such as salaries, legal, executive, and office real estate costs, also fall in this category.

Claims

- Claims represent the single largest expense for any insurer.
- They are the actual payouts made to policyholders to cover losses, alongside the administrative costs required to process and settle those losses.
- When looking at claims, you can examine the claims across different product lines and also look at the factors driving changes over time.

Prior Year Claims Development (PYD)

- It is the difference between prior year-end estimates of ultimate claim costs (excluding the effects of discounting and the risk adjustment for non-financial risk) and the current estimates for the same block of claims.
- A favourable development represents a reduction in the estimated ultimate claim costs during the period for that block of claims.
- Refer 05 Loss Reserving To learn more about the ladder method, which is used to calculate and project prior-year claims development.
